Bramwell Hydraulics

BH-250

250 t hydraulic press

Service Manual

Revision 3

For qualified maintenance personnel

1 Safety

1.1 General safety

Only qualified maintenance personnel who have read this manual may service the BH-250. Keep hands and tools out of the die space while the pump runs. Fit the safety blocks under the ram before you work beneath it. Wear safety glasses and oil-resistant gloves, and clean up spilled oil at once.

1.2 Lockout and tagout

Before you work on the hydraulic system, stop the pump, switch off the main isolator and fit your personal lock and tag. The system keeps stored hydraulic energy in the accumulator after the pump stops, so bleed the accumulator until its gauge reads 0 bar before you open any line. Lower the ram onto blocks and verify zero energy before you start.

2 Alarm table

The control shows these codes. Likely causes are listed in order of likelihood.

Code	Message	Severity	Likely causes	Action
E-200	Pump start	info	none	No action required.
E-217	Main cylinder pressure loss	critical	worn main cylinder seals (most likely) relief valve out of adjustment (possible)	See 4.1 Replace main cylinder seal kit (plant ref PROC-HP-04) See 4.3 Reset and test relief valve (plant ref PROC-HP-06)
E-221	Hydraulic oil temperature high	warning	contaminated hydraulic oil (most likely) relief valve out of adjustment (possible)	See 4.2 Change hydraulic oil and filter (plant ref PROC-HP-05) See 4.3 Reset and test relief valve (plant ref PROC-HP-06)
E-230	Pump pressure unstable	warning	contaminated hydraulic oil (most likely)	See 4.2 Change hydraulic oil and filter (plant ref PROC-HP-05)
E-245	Oil level low	warning	worn main cylinder seals (most likely)	See 4.1 Replace main cylinder seal kit (plant ref PROC-HP-04)
E-252	Relief valve pressure out of range	critical	relief valve out of adjustment (most likely)	See 4.3 Reset and test relief valve (plant ref PROC-HP-06)

3 Troubleshooting table

Find the symptom, then follow the remedy.

Symptom	Probable cause	Remedy
oil leaking under the ram, pressure dropping	worn main cylinder seals	See 4.1 Replace main cylinder seal kit (plant ref PROC-HP-04)
oil looks dark and milky, pump noisy	contaminated hydraulic oil	See 4.2 Change hydraulic oil and filter (plant ref PROC-HP-05)
press cannot reach tonnage, relief valve hissing	relief valve out of adjustment	See 4.3 Reset and test relief valve (plant ref PROC-HP-06)

4 Procedures

4.1 Replace main cylinder seal kit

Plant ref: PROC-HP-04 | Lockout required: yes | Time: 180 min

> **WARNING: Apply lockout and tagout before starting. Verify zero energy.**

1. Lower the ram onto blocks, stop the pump and apply lockout and tagout on the main isolator.
2. Bleed stored pressure from the accumulator until the gauge reads 0 bar.
3. Remove the cylinder gland and pull out the old seals.
4. Fit the SEAL-KIT-50 seals in the order shown in figure 4.
5. Refit the gland and top up the oil with OIL-HLP46-20L.
6. Remove the lockout, cycle the press 10 times at low pressure and check for leaks.

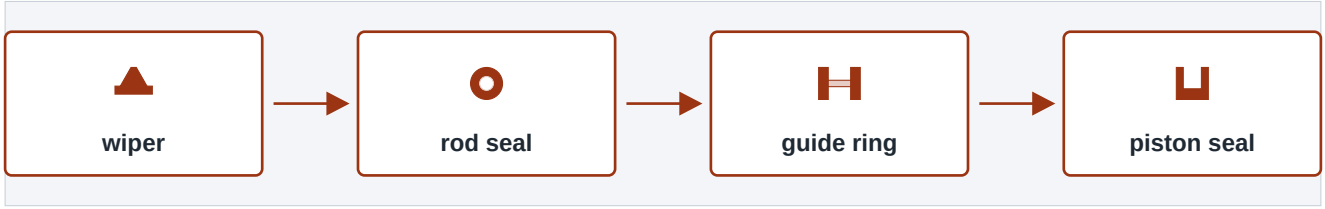

Figure 4: seal order

Order no.	Description	Qty
SEAL-KIT-50	SK-50 seal kit (order no. SEAL-KIT-50)	1
OIL-HLP46-20L	Hydraulic oil HLP 46, 20 litre pail	1

4.2 Change hydraulic oil and filter

Plant ref: PROC-HP-05 | Lockout required: yes | Time: 120 min

> **WARNING: Apply lockout and tagout before starting. Verify zero energy.**

1. Stop the pump, apply lockout and tagout, and bleed the accumulator to 0 bar.
2. Drain the tank and wipe out the sludge.
3. Replace the return filter with FLT-HYD-10.
4. Refill with OIL-HLP46-20L to the upper sight-glass mark.
5. Run the pump unloaded for 10 minutes and take an oil sample.

Order no.	Description	Qty
OIL-HLP46-20L	Hydraulic oil HLP 46, 20 litre pail	3
FLT-HYD-10	Hydraulic return filter 10 micron	1

4.3 Reset and test relief valve

Plant ref: PROC-HP-06 | Lockout required: yes | Time: 60 min

WARNING: Apply lockout and tagout before starting. Verify zero energy.

1. Stop the pump, apply lockout and tagout, and bleed the accumulator to 0 bar.
2. Remove the relief valve cap and replace the seal with SEAL-RV-25.
3. Remove the lockout and set the valve to 210 bar with the test gauge.
4. Lock the adjuster and record the setting.

Order no.	Description	Qty
SEAL-RV-25	Relief valve seal kit	1

5 Spare parts list

Order spare parts by part number.

Part number	Description	Fits
COUPLING-PMP-28	Pump coupling 28 mm	Hydraulic pump
FLT-HYD-10	Hydraulic return filter 10 micron	Hydraulic pump
GAUGE-PRS-250	Pressure gauge 0 to 250 bar	Pressure relief valve
HOSE-HYD-12	Hydraulic hose 12 mm, 1 m	Hydraulic pump
NUT-GLAND-50	Cylinder gland nut 50 mm	Main cylinder and seals
O-RING-NBR-12	O-ring NBR 12 mm, pack of 10	Pressure relief valve
OIL-HLP46-20L	Hydraulic oil HLP 46, 20 litre pail	Hydraulic pump Main cylinder and seals
SEAL-KIT-50	Main cylinder seal kit, 50 mm rod	Main cylinder and seals
SEAL-RV-25	Relief valve seal kit	Pressure relief valve
SENS-PT100-01	Temperature sensor PT100	Hydraulic pump
SWITCH-LIM-01	Limit switch	Main cylinder and seals
WIPER-CYL-50	Cylinder rod wiper 50 mm	Main cylinder and seals